

Coding & Solution

Date	30 September 2026
Team ID	SWTID-2026-5807
Project Name	AI-StudyBuddy
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python, JavaScript (React), HTML / CSS
Framework(s) Used	FastAPI, React.js, Tailwind CSS, MongoDB (Atlas)
Key Features Implemented	AI doubt-solving chatbot, notes / PDF summarizer, quiz and flashcard generator, personalized study planner with reminders, progress dashboard, and secure login (email / Google).
Pending / Incomplete Features	
Setup / Run Instructions	1) git clone <repo-url> 2) pip install -r requirements.txt (backend) and npm install (frontend) 3) copy .env.example to .env and add the MongoDB URI, LLM API key and JWT secret 4) uvicorn app.main:app --reload and npm start; open http://localhost:3000

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	
2	Meaningful variable and function names are used	
3	Code includes comments / documentation where necessary	
4	Error handling is implemented for critical operations	
5	The application runs without critical errors	
6	Code is committed to a version control repository	

Additional Notes / Comments:

The project follows a modular architecture with separate frontend and backend folders. API keys and secrets are stored in a .env file and are never committed to the repository. All AI features go through a single LLM client wrapper, so the model or provider can be changed easily. Unit tests and load tests are kept in the /tests folder.